

Final Exam Preparation: Level 0 - 2

Descriptions

Note: Questions are either a **Program** (requires `main`, `argc`, `argv`, and prints to standard output) or a **Function** (you only write the requested function and submit it without a `main`).

Level 0 & 1 (String Traversal & Basics)

- **aff_a** (Program): Takes a string. Prints the first 'a' it finds, then a newline. If no 'a' is found, just prints a newline. If no arguments are passed, prints 'a' followed by a newline.
- **aff_z** (Program): Ignores all arguments. Just prints 'z' followed by a newline.
- **ft_countdown** (Program): Displays all digits from 9 down to 0 in descending order, followed by a newline.
- **hello** (Program): Prints "Hello World!" followed by a newline.
- **maff_alpha** (Program): Prints the alphabet with alternating even letters in uppercase and odd letters in lowercase (aBcDeFgHiJkLmNoPqRsTuVwXyZ), followed by a newline.
- **only_z** (Program): Prints 'z' without a newline. (Often just 'z', check exact subject for newline).
- **first_word** (Program): Takes a string. Prints its first word (delimited by spaces or tabs), followed by a newline. Handles multiple spaces before the first word.
- **ft_putstr** (Function): Displays a string on the standard output.
- **ft_strcpy** (Function): Reproduces the behavior of the `strcpy` function (copies string `s2` into `s1`).
- **ft_strlen** (Function): Reproduces the behavior of the `strlen` function (returns the length of a string).
- **ft_swap** (Function): Swaps the contents of two integers whose addresses are passed as parameters.

- **repeat_alpha** (Program): Takes a string and prints it, but replaces each alphabetical character by itself repeated based on its alphabetical index ('a' prints once, 'b' prints twice, 'z' prints 26 times).
- **rev_print** (Program): Takes a string and displays it in reverse, followed by a newline.
- **rot_13** (Program): Takes a string and replaces every alphabetical character with the 13th character ahead in the alphabet, wrapping around 'z' to 'a'.
- **search_and_replace** (Program): Takes 3 arguments (a string, a letter, a letter). Replaces all instances of the first letter with the second letter in the string.
- **ulstr** (Program): Takes a string and reverses the case of all its letters (uppercase becomes lowercase and vice versa).

Level 2 (Advanced Strings, Bitwise, & Logic)

- **alpha_mirror** (Program): Takes a string. Replaces every alphabetical character with its opposite alphabetical character ('a' becomes 'z', 'b' becomes 'y', 'M' becomes 'N').
- **camel_to_snake** (Program): Takes a string in `camelCase` and converts it into `snake_case` (e.g., "helloWorld" -> "hello_world").
- **do_op** (Program): Takes two numbers and an operator as strings (e.g., "5", "*", "4"). Computes the result and prints it.
- **ft_atoi** (Function): Reproduces the behavior of the standard `atoi` function (converts a string to an integer, handling leading whitespaces and a single '+' or '-').
- **ft_strcmp** (Function): Reproduces the behavior of the `strcmp` function (compares two strings and returns the ASCII difference of the first unmatched character).
- **ft_strdup** (Function): Reproduces the behavior of the `strdup` function (allocates memory with `malloc` and duplicates the given string into it).
- **ft_strrev** (Function): Reverses a string *in place* (modifies the original string directly) and returns a pointer to it.
- **inter** (Program): Takes two strings. Prints characters that appear in *both* strings, in the exact order they appear in the first string. No duplicates are printed.

- **is_power_of_2** (Function): Takes an unsigned int. Returns 1 if it is a power of 2, otherwise returns 0.
- **last_word** (Program): Takes a string. Prints its last word (delimited by spaces or tabs), followed by a newline.
- **max** (Function): Takes an array of integers and its length. Returns the largest number in the array.
- **print_bits** (Function): Takes an unsigned char and prints its binary representation to standard output (e.g., passing 2 prints "00000010").
- **reverse_bits** (Function): Takes an unsigned char, reverses its bits (e.g., 01000001 becomes 10000010), and returns the new byte.
- **snake_to_camel** (Program): Takes a string in snake_case and converts it into camelCase (e.g., "hello_world" -> "helloWorld").
- **swap_bits** (Function): Takes an unsigned char and swaps its two halves (e.g., 0100 0001 becomes 0001 0100).
- **union** (Program): Takes two strings. Prints all characters that appear in *either* string, in the exact order they appear (checking string 1 first, then string 2). No duplicates are printed.
- **wdmatch** (Program): Takes two strings. Checks if all characters from the first string are "hidden" inside the second string in the exact same order. If they are, it prints the first string.